Put S, P or B next to each of the following adaptations to classify them as Structural, Physiological or Behavioural.

Desert Plants and Animals

Camels - How have they adapted to their environment?



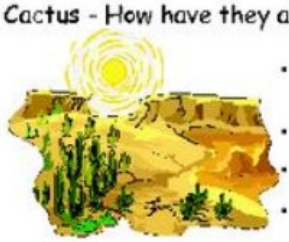
- Long eye lashes to keep sand out of their eyes.
- Nostrils can close to keep sand out.
- Being able to go for a week without water.
- Wide feet for walking in the sand.
- Store fat in their humps.

Rattlesnake - How have they adapted to their environment?



- Re-grow fangs when they get broken.
- It's colour ranges depending on its habitat.
- Has heat-sensing facial pits to find warm blooded prey.
- When threatened it will coil, rattle fearsomely and stand its ground.

Cactus - How have they adapted to their environment?



- Sharp spines to protect them from herbivorous animals.
- A waxy coating to reduce evaporation of moisture.
- Stores lots of water when it rains.
- Flowers open at night and close up in the day.
- Tiny hairs soak up water from precipitation.

Read the information on the next page and write the adaptations in their correct category.

Structural	Physiological	Behavioural

Plant Adaptations in the Tropical Rainforest Biome

The bark on the rainforest trees is thin which prevents other plants from growing on them and supports evaporation from the excess humidity. The leaves on most of the plants have what is called drip tips which help plants shed the excess moisture from the rainfall. The trees have long, very straight trunks and don't branch out until they reach the canopy where sunlight can be found. Leaves of plants and trees are very large to help them absorb as much of the sunlight as is possible. Some of the tropical Rainforest plants are actually carnivorous, which means they eat meat. They have an opening filled with nectar that will attract insects, such as ants and flies. Many flowering plants will grow on to trees in able to get sunshine (epiphytes).

Animal Adaptations in the Tropical Rainforest Biome

Animals come in various colors which act as a camouflage to protect them from their predators. Many of the animals are able to move freely among the many trees which act as both food and shelter for them. Quite a few of the tropical rainforest animals are able to swim and cope with a wet and swamp like environment. Some animals (bats) are nocturnal which help them avoid the warmer daytime temperatures. Many of the birds have larger and stronger beaks. Some animals are poisonous which protect them from their predators. Due to the competition for food with so many animals, some animals are very specialized and only eat one type of plant or insect.

Plant Adaptations in the Tundra Biome

Plants in the Tundra have adapted in a variety of ways. They grow close together, low to the ground and they remain small. Many plants in the biome have a wax type of fuzzy, hairy coating on them which helps to shield them from the cold and the wind. This coating also helps them to retain heat and moisture and it protects the plant seeds to allow for reproduction. Plants found here will typically have small leaves which helps the plants to retain moisture. Only the top layer of soil thaws out in the Tundra, below that is the permafrost, therefore the plants have very shallow root systems. Plants like lichens and moss can survive on bare rock with a bit of moisture. Most of the plants in the Tundra Biome are perennials and don't die off in the winter, they have long life cycles to help with the short growing season.

Animal Adaptations in the Tundra Biome

Animals have many adaptations to survive in this harsh environment. Animals need shelter and insulation in the Tundra. The animals here tend to have thicker and warmer feathers and fur. Many of them have larger bodies and shorter arms, legs and tails which helps them retain their heat better and prevent heat loss. Many of the birds of the tundra have two coats of feathers to help keep them warm. Many animals of the Tundra have feet that are lined with fur to help keep them warm. Some of the animals of the Tundra (bears, marmot, arctic squirrels) will hibernate for the winter and others will burrow (lemmings, ermine). Many of the insects of the Tundra will spend their entire life buried in the soil, rocks or plants which acts as a shelter for them.

Things needed for Natural Selection

Read the text below and fill in the table.

Some red-bellied black snakes are longer than others. Head size also varies and limits the size of the prey the snake can eat. Snakes sometimes eat cane toads, which are toxic. Larger cane toads are more toxic than smaller ones. Snakes with larger bodies can cope with higher levels of toxin. Red-bellied black snakes lay up to 18 live babies at a time, but many won't survive to reproduce. Because longer snakes with smaller heads survive better in areas with cane toads, they are able to reproduce more. The longer length and smaller head traits are passed on to their offspring and become more common in the population.

Letter	Word	Meaning	Part from the example which matches
D		Animals have different traits, originally would have been caused by mutations	
I		Genes are passed onto offspring	
S		Things in the environment that organisms compete for or can affect survival	
C		Not all animals can survive so only the fittest survive and pass on genes	

Match the word to the correct definition and put it in the box.

Mutation, Evolution, Adaptation, Charles Darwin, Diversity, Fitness, Natural selection

Differences between and within species	
Scientist who developed the theory of evolution and 'survival of the fittest'	
The **process** by which evolution happens and species adapt	
A trait that helps an animal survive or reproduce in it's habitat	
When an organism survives and reproduces, the higher number of offspring the higher their …	
Change over time as animals adapt to their environment	
A sudden change in the DNA of an organism	

Evidence for evolution

Fill in the following table:

Type of evidence	How it gives us evidence for evolution:	Example:
Fossils		
DNA/Genetics		
Embryology		
Homologous structures		